WORKED SOLUTION

STEP I 2012, Question 12

Probability and Statistics • Pdf

STEP I 2012, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q12

The key idea

Everything depends on which year the extinguisher becomes faulty in, so first turn the density into a cumulative distribution function and read off the probabilities of the fault arriving in the first, second and third year. Once we know the year, the tests are just a run of independent passes followed by a fail. The final part is a conditional probability, and the event "faulty at 18 months" is again a statement about which interval T lies in.

Solution

The distribution of T

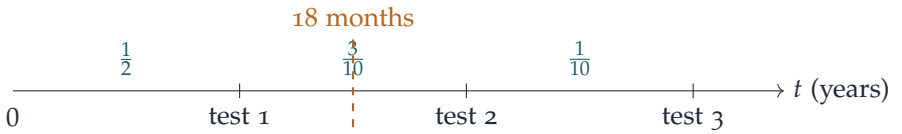
For $t \geq 0$, substituting $u = 1 + s^2$ (so $\mathrm{d}u = 2s\,\mathrm{d}s$),

$$F(t) = \mathrm{P}(T \leq t) = \int_0^t \frac{2s}{(1+s^2)^2}\,\mathrm{d}s = \left[-\frac{1}{1+s^2}\right]_0^t = 1 - \frac{1}{1+t^2} = \frac{t^2}{1+t^2}.$$

So $F(1) = \frac{1}{2}$, $F(2) = \frac{4}{5}$, $F(3) = \frac{9}{10}$, and

$$\mathrm{P}(0 < T < 1) = \tfrac{1}{2}, \qquad \mathrm{P}(1 < T < 2) = \tfrac{4}{5} - \tfrac{1}{2} = \tfrac{3}{10}, \qquad \mathrm{P}(2 < T < 3) = \tfrac{9}{10} - \tfrac{4}{5} = \tfrac{1}{10}.$$

Since T is continuous, the probability that the fault arrives exactly on a test day is zero, so we needn't worry about whether $<$ or $\leq$ is used at the endpoints.



Destroyed exactly three years after manufacture

Let D be the event that the extinguisher is destroyed at the test three years after manufacture. For this to happen it must be faulty by the third test (so $T < 3$), it must

pass every test it takes while faulty before then, and it must fail the third test. A non-faulty extinguisher always passes, so only the tests after time T matter, and they are independent with failure probability p.

- If $0 < T < 1$: it is faulty at tests 1, 2 and 3, so it must pass, pass, fail: probability $(1-p)^2 p$.
- If $1 < T < 2$: it passes test 1 automatically, then must pass test 2 and fail test 3: probability $(1-p)p$.
- If $2 < T < 3$: it passes tests 1 and 2 automatically and must fail test 3: probability p.

If $T > 3$ it is not faulty at test 3 and so cannot be destroyed then. By the law of total probability,

$$\begin{aligned} \mathrm{P}(D) &= \tfrac{1}{2}(1-p)^2 p + \tfrac{3}{10}(1-p)p + \tfrac{1}{10}p \\ &= \frac{p}{10}\Big(5(1-2p+p^2) + 3(1-p) + 1\Big) \\ &= \frac{p\,(5p^2 - 13p + 9)}{10}, \end{aligned}$$

as required.

Faulty at 18 months, given destroyed at three years

Let E be the event that the extinguisher was faulty 18 months after manufacture, that is, $T < \frac{3}{2}$. We want

$$\mathrm{P}(E \mid D) = \frac{\mathrm{P}(E \cap D)}{\mathrm{P}(D)}.$$

Ask yourself The interval $(0, \frac{3}{2})$ straddles a test. How do we split it?

What matters for the tests is the year in which the fault arrives, so split E into $0 < T < 1$ and $1 < T < \frac{3}{2}$. We have $F(\frac{3}{2}) = \dfrac{9/4}{13/4} = \dfrac{9}{13}$, so

$$\mathrm{P}\big(1 < T < \tfrac{3}{2}\big) = \tfrac{9}{13} - \tfrac{1}{2} = \tfrac{5}{26}.$$

Given $0 < T < 1$, the chance of D is $(1-p)^2 p$ as before; given $1 < T < \frac{3}{2}$, the fault still arrives in the second year, so the chance of D is $(1-p)p$. Hence

$$\begin{aligned} \mathrm{P}(E \cap D) &= \tfrac{1}{2}(1-p)^2 p + \tfrac{5}{26}(1-p)p \\ &= \frac{p(1-p)}{26}\big(13(1-p) + 5\big) = \frac{p(1-p)(18-13p)}{26}. \end{aligned}$$

Dividing by $\mathrm{P}(D)$ (which is positive for $0 < p \leq 1$, because $5p^2 - 13p + 9$ has discriminant $169 - 180 < 0$ and so is always positive), the factor p cancels and

$$\mathrm{P}(E \mid D) = \frac{10(1-p)(18-13p)}{26(5p^2 - 13p + 9)} = \frac{5(1-p)(18-13p)}{13(5p^2 - 13p + 9)}.$$

Checks

- When $p = 1$ every faulty extinguisher is destroyed at its first test after the fault, so destruction at year 3 means the fault arrived in $(2, 3)$: $\mathrm{P}(D) = \frac{1}{10}$ and $\mathrm{P}(E \mid D) = 0$. Both formulae agree.
- As $p \to 0$, $\mathrm{P}(E \mid D) \to \dfrac{5 \cdot 18}{13 \cdot 9} = \dfrac{10}{13}$, which is $\mathrm{P}(T < \frac{3}{2} \mid T < 3) = \dfrac{9/13}{9/10}$: when the test barely detects faults, the destruction tells us almost nothing beyond $T < 3$.
- A simulation of 10^6 extinguishers with $p = 0.4$ gave $\mathrm{P}(D) \approx 0.1838$ (exact 0.184) and $\mathrm{P}(E \mid D) \approx 0.642$ (exact $\frac{192}{299} \approx 0.642$).

Common slips

- Forgetting that a faulty extinguisher has to *pass* the earlier tests before failing the third one.
- Using $\mathrm{P}(T < \frac{3}{2}) = \frac{9}{13}$ as a single block in the final part: faults in the first year and in the first half of the second year face different numbers of tests.

Beyond STEP

Here T is a continuous lifetime, but we only observe it through discrete, imperfect inspections. This is the setting of *interval-censored* data in survival analysis and reliability theory, and the last part is a small Bayesian update: the prior distribution of T is revised in the light of an observed failure. Grimmett and Welsh, *Probability: An Introduction*, covers continuous distributions and conditioning of this kind.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.